Project: Worked examples
Title: Mechanism kinematics
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1. Scope and method

Calculates the position, velocity, acceleration, input torque and forces of a one-degree-of-freedom planar four-bar, slider-crank or cam mechanism. The input turns at constant speed, and the links are rigid, with no clearance and no friction.

WHAT IT CALCULATES	RELATION
Four-bar position	$a\cdot e^{j\theta_2} + b\cdot e^{j\theta_3} = d + c\cdot e^{j\theta_4}$, with $\ell = \lvert A - O_4\rvert$, $\cos\gamma = (c^2 + \ell^2 - b^2)/(2c\ell)$ and $\theta_4 = \varphi \mp \gamma$ (φ the direction $O_4 \to A$; open: −, crossed: +); θ_3 from B − A
Four-bar velocity and acceleration	differentiated loop, 2×2 system with $\det = b\cdot c\cdot\sin(\theta_3 - \theta_4)$; $\omega_3 = \omega_2\cdot a\cdot\sin(\theta_4 - \theta_2)/(b\cdot\sin(\theta_3 - \theta_4))$, $\omega_4 = \omega_2\cdot a\cdot\sin(\theta_3 - \theta_2)/(c\cdot\sin(\theta_3 - \theta_4))$; α from the second derivative at constant ω_2
Grashof class and transmission angle	$s + l < p + q$ (crank-rocker, double-crank or double-rocker by the shortest link), = at the change point, > non-Grashof; transmission angle $\cos\mu = (b^2 + c^2 - \ell^2)/(2bc)$ folded to 0 … 90°; toggle where $\ell = b + c$ or $\lvert b - c\rvert$
Slider-crank position	$x = r\cdot\cos\theta + \sigma\cdot\sqrt{(l^2 - (r\cdot\sin\theta - e)^2)}$ (σ = +1 open, −1 crossed); $dx/d\theta = -r\cdot\sin\theta - r\cdot\cos\theta\cdot(r\cdot\sin\theta - e)/(x - r\cdot\cos\theta)$; dead centres where crank and rod are collinear; $\mu = 90° - \lvert$rod angle to the slide line$\rvert$
Input torque and forces	$T_{in}\cdot\omega_{in} = \Sigma F\cdot v$ by virtual work; four-bar $T_{in} = (T_L + J\cdot\alpha_4)\cdot\omega_4/\omega_2$, tangential pin force $P = (T_L + J\cdot\alpha_4)/c$ and coupler force $P/\sin\mu$; slider $T_{in} = (m\cdot a + F_L)\cdot dx/d\theta$, slide-direction force $F_x = m\cdot a + F_L$, rod force $F_x/\cos\theta_3$ and guide reaction $F_x\cdot\tan\theta_3$; RMS torque and mean power over the sweep by Simpson's rule
Assembly limits and loop closure	four-bar toggles at $\cos\theta_2 = (d^2 + a^2 - \ell^2)/(2ad)$, $\ell = b + c$ or $\lvert b - c\rvert$; slider $\sin\theta = (e \pm l)/r$; the sweep is cut at them and the unbounded peaks are not reported; closure $\lvert a\cdot e^{j\theta_2} + b\cdot e^{j\theta_3} - c\cdot e^{j\theta_4} - d\rvert$ (slider: $\lvert r\cdot e^{j\theta} + l\cdot e^{j\theta_3} - (x + j\cdot e)\rvert$) below 1e-9 mm at every step
Cam motion	per segment with $x = (\theta - \theta_o)/\beta$, rise $s = s_o + h\cdot f(x)$, return $s = s_o - h\cdot f(x)$; cycloidal $f = x - \sin(2\pi x)/(2\pi)$; 3-4-5 $f = 10x^3 - 15x^4 + 6x^5$; 4-5-6-7 $f = 35x^4 - 84x^5 + 70x^6 - 20x^7$; modified trapezoid from its acceleration pieces ($C_v = 2$, $C_a = 8\pi/(\pi + 2)$, $C_j = 32\pi^2/(\pi + 2)$); SVAJ at constant ω: $v = \omega\cdot s'$, $a = \omega^2\cdot s''$, $j = \omega^3\cdot s'''$ ($' = d/d\theta$), so $v_{pk} = C_v\cdot h\cdot\omega/\beta$

WHAT IT CALCULATES	RELATION
Cam contact	roller pressure angle $\varphi = \arctan[(s' - e)/(\sqrt{(R_p^2 - e^2)} + s)]$ with $R_p = R_b + R_r$, base circle $R_{p,req} = \sqrt{(s_o,req^2 + e^2)}$ with $s_o,req = \max(\lvert s' - e\rvert/\tan \varphi_{max} - s)$; pitch-curve radius $\rho = \lvert P'\rvert^3/(P' \times P'')$, undercut where $\rho_{min} \le R_r$; flat-faced profile radius $\rho = R_b + s + s''$ above zero; follower force $F = m \cdot a + k \cdot s + F_{pre} + F_{ext}$ (the follower leaves the cam where $F < 0$), input torque $T = F \cdot s'$, r...

Basis: Position, velocity and acceleration analysis; transmission angle; Grashof condition[1](Ch. 4, 6, 7, 6th ed.); Loop-closure position analysis; velocity and acceleration of planar linkages[2](Ch. 4–6, 5th ed.).

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
mode	Kind of mechanism	Cam		spec: Indexing station IDX-12 §2 rev A	any
e_{off}	Offset e of the slide line above the crank centre, or of the cam follower axis at +x: positive e reduces the pressure angle where ds/dθ exceeds e (optional; blank = 0)	10	mm	drawing: IDX-12-CAM rev A	-1000 ... 1000 mm
steps	Intervals N of the table (N + 1 rows, a whole number, at most 3600; the peaks do not depend on it)	72		assumption: 5° table step.	4 ... 720
n_{in}	Input speed (constant)	120	rpm	spec: Indexing station IDX-12 §2: 120 index cycles/min rev A	0.1 ... 30000 rpm
follower	Follower type: roller (translating, pressure angle and undercut checked) or flat-faced (in-line)	Roller		drawing: IDX-12-CAM rev A	any
Rb	Base-circle radius of the cam profile	40	mm	drawing: IDX-12-CAM rev A	1 ... 1000 mm
R_r	Roller radius (roller follower)	8	mm	drawing: IDX-12-CAM rev A	1 ... 200 mm
φ_{max}	Maximum pressure angle allowed (roller follower)	30	°	assumption: Usual limit for a translating roller follower; set it from your design rule.	5 ... 45 °

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
segs	Cam segments: rise, dwell or return, with motion law, angle and lift; the angles add to 360° and the rises equal the returns	4 row(s)		spec: Indexing station IDX-12 §2.3 motion table rev A	any
m_f	Follower mass (all moving parts that follow the cam)	0.2	kg	assumption: Example: follower arm, roller and tooling; take it from the CAD mass properties.	0 … 1000 kg
k_{sp}	Return spring rate	5	kN/m	assumption: Example return-spring rate; link the rate k of the spring sheet.	0 … 1000 kN/m
F_{pre}	Spring preload at the base circle	20	N	assumption: Example preload at the base circle; link a force of the spring sheet.	0 … 100000 N
F_{ext}	External load on the follower, positive against the rise	0	N	assumption: No external load on the follower in this example.	-100000 … 100000 N

TABLE ROW	VALUES
segs[1]	kind = Rise; law = Cycloidal; beta = 120°; h = 20 mm
segs[2]	kind = Dwell; law = —; beta = 60°; h = 0 mm
segs[3]	kind = Return; law = 3-4-5 polynomial; beta = 120°; h = 20 mm
segs[4]	kind = Dwell; law = —; beta = 60°; h = 0 mm

3. Assumptions

A1. The mechanism is rigid and planar, has one degree of freedom, and has no joint clearance or friction. (Scope limits of the model.)
A2. The input turns at a constant speed. (Scope limits of the model.)
A3. The position is the closed-form solution of the vector loop, on the chosen circuit. (Open and crossed are the two branches: $\theta_4 = \varphi - \gamma$ and $\varphi + \gamma$ for the four-bar, $\sigma = +1$ and -1 for the slider.)
A4. The sweep stays on the chosen circuit and follows it by continuity through a change point. (Only a change point, with all four links collinear ($\theta_2 = 0$ or π), flips the branch.)
A5. A linkage with one change point per turn repeats every 720°, and a sweep of more than one turn warns. (A kite, for example 40/40/100/100, is such a linkage.)
A6. The table leaves out angles where the linkage cannot be assembled or sits at a toggle. (It counts them, and at a toggle the velocity system is singular.)
A7. The input torque comes from the output load and the inertia of the output element only. (The crank and the coupler or connecting rod are massless: add their mass to the output element for a first allowance.)
A8. The crank, the coupler and the connecting rod are massless two-force members. (Rigid body, frictionless.)
A9. The peaks are those of the continuous motion over the swept range, not of the printed table. (A dense

scan refined by golden section finds them, so they do not depend on the step count of the table (up to 3600).)
A10. At an assembly limit the sweep is cut, and the unbounded peaks, torque and forces are not reported. (The limit angles are listed: $\ell = b + c$ or $|b - c|$ (four-bar), $|r \cdot \sin\theta - e| = l$ (slider).)
A11. A change-point four-bar is continued through its singular position on the same branch. (Its speeds and torques next to the singular position are not reported.)
A12. The cam drives a translating follower at constant speed. (Norton, Design of Machinery, cam design: the four classical laws, pressure angle and radius of curvature.)
A13. The follower force is $m \cdot a + k \cdot s + F_{pre} + F_{ext}$, and contact is lost below zero. (The spring rate and the preload are those at the base circle, and the segment table must add to 360° and close.)
A14. Pressure angle and undercut are checked for the roller follower, and the profile radius for the flat-faced one. (The roller has the prime radius $R_p = R_b + R_r$.)
A15. The output speed is a kinematic proxy at constant input speed, not the speed of an actuated cylinder. (For motor sizing, use the peak input_{torque} for the peak check and $T_{rms,in}$ with n_{in} for the thermal sizing. The mean power is P_{mean}.)
A16. steps (Intervals N of the table (N + 1 rows, a whole number, at most 3600; the peaks do not depend on it)) is taken as 72. (5° table step.)
A17. φ_{max} (Maximum pressure angle allowed (roller follower)) is taken as 30 °. (Usual limit for a translating roller follower; set it from your design rule.)
A18. m_f (Follower mass (all moving parts that follow the cam)) is taken as 0.2 kg. (Example: follower arm, roller and tooling; take it from the CAD mass properties.)
A19. k_{sp} (Return spring rate) is taken as 5 kN/m. (Example return-spring rate; link the rate k of the spring sheet.)
A20. F_{pre} (Spring preload at the base circle) is taken as 20 N. (Example preload at the base circle; link a force of the spring sheet.)
A21. F_{ext} (External load on the follower, positive against the rise) is taken as 0 N. (No external load on the follower in this example.)

4. Working

[15] $\theta_{sw} = \theta$ [3](kinematics)
= 73 rows, θ from 0° to 360°
= [0, 5, 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, … (73 items)] deg
Table: cam angle

[16] $s_{sw} = s = s_0 + h \cdot f(\theta/\beta)$ [3](kinematics)
= 73 rows, θ from 0° to 360°
= [0, 0.0094867, 0.075117, 0.24921, 0.57669, 1.092, 1.8169, 2.7587, 3.91, 5.2492, 6.7418, 8.3 items)] mm
Table: follower displacement

[17] $v_{sw} = v = \omega \cdot ds/d\theta$ [3](kinematics)
= 73 rows, θ from 0° to 360°
= [0, 0.0040889, 0.016077, 0.035147, 0.06, 0.088942, 0.12, 0.15106, 0.18, 0.20485, 0.22392, 0 (73 items)] m/s
Table: follower velocity

[18] $\alpha_{sw} = \alpha = \omega^2 \cdot d^2s/d\theta^2$ [3](kinematics)
= 73 rows, θ from 0° to 360°
= [0, 1.1709, 2.2619, 3.1989, 3.9178, 4.3697, 4.5239, 4.3697, 3.9178, 3.1989, 2.2619, 1.1709, m/s²
Table: follower acceleration

[19] $j_{sw} = j = \omega^3 \cdot d^3s/d\theta^3$ [3](kinematics)
= 73 rows, θ from 0° to 360°
= [170.55, 164.74, 147.7, 120.59, 85.273, 44.141, 1.0443e-14, -44.141, -85.273, -120.59, -147 (73 items)] m/s³
Table: follower jerk

[20] $\varphi_{sw} = \varphi = \arctan[(s' - e)/(\sqrt{(R_p^2 - e^2)} + s)]$ [3](kinematics)
= 73 rows, θ from 0° to 360°
= [-12.025, -11.642, -10.507, -8.6775, -6.2746, -3.4811, -0.52955, 2.3281, 4.8597, 6.8841, 8.
… (73 items)] deg
Table: pressure angle

[21] $\kappa_{sw} = \kappa = 1/\rho$ of the pitch curve (positive convex, negative concave) [3](kinematics)
= 73 rows, θ from 0° to 360°
= [0.020833, 0.01767, 0.01465, 0.011975, 0.0098989, 0.0086679, 0.008423, 0.0091267, 0.010569,
0.012452, 0.014489, 0.016465, … (73 items)] 1/mm
Table: curvature of the pitch curve (the radius of curvature is 1/κ where κ is positive)

[22] $F_{sw} = F = m \cdot a + k \cdot s + F_{pre} + F_{ext}$ [3](kinematics)
= 73 rows, θ from 0° to 360°
= [20, 20.282, 20.828, 21.886, 23.667, 26.334, 29.989, 34.667, 40.334, 46.886, 54.161, 61.948
items)] N
Table: follower force on the cam

[23] $T_{in,sw} = T = F \cdot ds/d\theta$ [4](work)
= 73 rows, θ from 0° to 360°
= [0, 0.0065993, 0.026647, 0.061213, 0.113, 0.18639, 0.28638, 0.41673, 0.57774, 0.76432, 0.96
… (73 items)] N·m
Table: input torque by virtual work

[24] seg_{start} = Σ β of the segments before [3](kinematics)
= 4 segments
= [0, 120, 180, 300] deg
Segments: start angle

[25] seg_{beta} = β [3](kinematics)
= 4 segments
= [120, 60, 120, 60] deg
Segments: angle

[26] $seg_{s,end}$ = s at the end of the segment [3](kinematics)
= 4 segments
= [20, 20, 0, 0] mm
Segments: displacement at the end

[27] $seg_{v,pk}$ = max |v| in the segment [3](kinematics)
= 4 segments
= [0.24, 0, 0.225, 0] m/s
Segments: peak velocity

[28] $seg_{a,pk}$ = max |a| in the segment [3](kinematics)
= 4 segments
= [4.5239, 0, 4.1569, 0] m/s²
Segments: peak acceleration

[29] $seg_{j,pk}$ = max |j| in the segment [3](kinematics)
= 4 segments
= [170.55, 0, 259.2, 0] m/s³
Segments: peak jerk

[30] lift = max s [3](kinematics)
= over the revolution
= 20 mm
Greatest follower displacement

[31] v_{pk} = max |v| [3](kinematics)
= scan of each segment, refined
= 0.24 m/s
Peak follower velocity

[32] a_{pk} = max |a| [3](kinematics)
= scan of each segment, refined
= 4.5239 m/s²
Peak follower acceleration
[33] j_{pk} = max |j| [3](kinematics)
= scan of each segment, refined
= 259.2 m/s³
Peak follower jerk
[34] F_{max} = max F [3](kinematics)
= scan of each segment, refined
= 120 N
Greatest follower force on the cam
[35] F_{min} = min F [3](kinematics)
= scan of each segment, refined
= 20 N
Smallest follower force on the cam (negative: the follower leaves the cam)
[36] $\varphi_{max,pk}$ = max |φ| [3](kinematics)
= scan of each segment, refined
= 26.596 deg
Largest pressure angle
[37] $Rb_{min,phi}$ = R_b that keeps φ at its limit [3](kinematics)
= s_0 = max(|s' − e|/tan φ_{max} − s), R_p = √(s_0^2 + e^2)
= 32.963 mm
Smallest base-circle radius that keeps the pressure angle within its limit (roller follower)
[38] ρ_{min} = 1/κ_{max} [3](kinematics)
= pitch curve
= 44.681 mm
Smallest radius of curvature of the pitch curve
[39] u_{rho} = R_r / ρ_{min} [3](kinematics)
= 8 mm / 44.681 mm
= 0.17905
Roller radius against the smallest radius of curvature (undercut at 1 or more)
[40] u_{phi} = φ_{max} / $\varphi_{allowed}$ [3](kinematics)
= 26.596° / 30°
= 0.88653
Largest pressure angle against the allowed one
[41] $input_{torque}$ = max |F·ds/dθ| [4](work)
= virtual work, scan of each segment, refined
= 1.5345 N·m
Peak input torque by virtual work
[42] $output_{motion}$ = max |v| [3](kinematics)
= follower
= 0.24 m/s
Peak follower velocity
[43] $peak_{accel}$ = max |a| [3](kinematics)
= follower
= 4.5239 m/s²
Peak follower acceleration
[44] $follower_{force}$ = max |F| [3](kinematics)
= m·a + k·s + F_{pre} + F_{ext}
= 120 N
Peak axial force of the follower on the cam

[45] $F_{pin,pk} = \max |F| / \cos \varphi$ [3](kinematics)
= roller contact normal force
= 121.33 N
Peak contact normal force on the cam, the load on the cam shaft's bearings

[46] $F_{spring,max} = k \cdot h + F_{pre}$ [3](kinematics)
= spring force at full lift, no inertia
= 120 N
Spring force at full lift

[47] $T_{rms,in} = \sqrt{(\text{mean } T^2)}$ [4](work)
= Simpson per segment
= 0.69759 N·m
RMS input torque over the cam revolution

[48] $P_{mean} = \omega \cdot \text{mean}(T)$ [4](work)
= Simpson per segment
= -4.375e-13 kW
Mean input power over the cam revolution

[49] $f_{in} = n_{in} / 60$
= rpm to Hz
= 2 Hz
Cam-shaft frequency, for a spring surge check

5. Results

SYMBOL	VALUE	DESCRIPTION
$input_{torque}$	1.534 N·m	Peak input torque by virtual work
$output_{motion}$	0.2400 m/s	Peak follower velocity
$peak_{accel}$	4.524 m/s²	Peak follower acceleration
$follower_{force}$	120 N	Peak axial force of the follower on the cam
$F_{pin,pk}$	121.3 N	Peak contact normal force on the cam, the load on the cam shaft's bearings
$T_{rms,in}$	0.6976 N·m	RMS input torque over the cam revolution
n_{in}	120 rpm	Cam-shaft speed, passed through
f_{in}	2 Hz	Cam-shaft frequency, for a spring surge check
P_{mean}	-4.375e-13 kW	Mean input power over the cam revolution
lift	20 mm	Greatest follower displacement
$F_{spring,max}$	120 N	Spring force at full lift
θ_{sw}	[0, 5, 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, … (73 items)] deg	Table: cam angle
s_{sw}	[0, 0.0094867, 0.075117, 0.24921, 0.57669, 1.092, 1.8169, 2.7587, 3.91, 5.2492, 6.7418, 8.3428, … (73 items)] mm	Table: follower displacement

SYMBOL	VALUE	DESCRIPTION
v_{sw}	[0, 0.0040889, 0.016077, 0.035147, 0.06, 0.088942, 0.12, 0.15106, 0.18, 0.20485, 0.22392, 0.23591, … (73 items)] m/s	Table: follower velocity
a_{sw}	[0, 1.1709, 2.2619, 3.1989, 3.9178, 4.3697, 4.5239, 4.3697, 3.9178, 3.1989, 2.2619, 1.1709, … (73 items)] m/s²	Table: follower acceleration
j_{sw}	[170.55, 164.74, 147.7, 120.59, 85.273, 44.141, 1.0443e-14, -44.141, -85.273, -120.59, -147.7, -164.74, … (73 items)] m/s³	Table: follower jerk
φ_{sw}	[-12.025, -11.642, -10.507, -8.6775, -6.2746, -3.4811, -0.52955, 2.3281, 4.8597, 6.8841, 8.2863, 9.0164, … (73 items)] deg	Table: pressure angle
κ_{sw}	[0.020833, 0.01767, 0.01465, 0.011975, 0.0098989, 0.0086679, 0.008423, 0.0091267, 0.010569, 0.012452, 0.014489, 0.016465, … (73 items)] 1/mm	Table: curvature of the pitch curve (the radius of curvature is 1/κ where κ is positive)
F_{sw}	[20, 20.282, 20.828, 21.886, 23.667, 26.334, 29.989, 34.667, 40.334, 46.886, 54.161, 61.948, … (73 items)] N	Table: follower force on the cam
$T_{in,sw}$	[0, 0.0065993, 0.026647, 0.061213, 0.113, 0.18639, 0.28638, 0.41673, 0.57774, 0.76432, 0.96511, 1.163, … (73 items)] N·m	Table: input torque by virtual work
seg_{start}	[0, 120, 180, 300] deg	Segments: start angle
seg_{beta}	[120, 60, 120, 60] deg	Segments: angle
$seg_{s,end}$	[20, 20, 0, 0] mm	Segments: displacement at the end
$seg_{v,pk}$	[0.24, 0, 0.225, 0] m/s	Segments: peak velocity
$seg_{a,pk}$	[4.5239, 0, 4.1569, 0] m/s²	Segments: peak acceleration

SYMBOL	VALUE	DESCRIPTION
$seg_{j,pk}$	[170.55, 0, 259.2, 0] m/s³	Segments: peak jerk
seg_{law}	"1: rise (Cycloidal); 2: dwell; 3: return (3-4-5 polynomial); 4: dwell"	Segments: kind and motion law
v_{pk}	0.2400 m/s	Peak follower velocity
a_{pk}	4.524 m/s²	Peak follower acceleration
j_{pk}	259.2 m/s³	Peak follower jerk
F_{max}	120 N	Greatest follower force on the cam
F_{min}	20 N	Smallest follower force on the cam (negative: the follower leaves the cam)
$\varphi_{max,pk}$	26.60°	Largest pressure angle
$Rb_{min,phi}$	32.96 mm	Smallest base-circle radius that keeps the pressure angle within its limit (roller follower)
ρ_{min}	44.68 mm	Smallest radius of curvature of the pitch curve
u_{rho}	0.18	Roller radius against the smallest radius of curvature
u_{phi}	0.89	Largest pressure angle against the allowed one (governing)

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
PASS	Largest pressure angle against the allowed pressure angle	26.60°	30°	0.89	[3](kinematics)
PASS	Smallest radius of curvature against the roller radius	ρ_{min} = 44.681 mm	> 8 mm	—	[3](kinematics)
PASS	Smallest follower force against zero	F_{min} = 20 N	≥ 0 N	—	[3](kinematics)

7. Applicability

None.

8. Verification

Template mech.mechanism@1.3.0 Content hash 2f3baaba1b0ea215 Engine calc-engine/1.0.0
Known-answer cases passed 16/16 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0

9. References

1. Norton, Design of Machinery.
2. Budynas RG, Nisbett JK. Shigley's Mechanical Engineering Design. 11th ed. McGraw-Hill Education; 2020.
3. Cam follower.
4. Virtual.